

Monolithic electro-optic platform on silicon with bandwidth of 100 GHz and beyond

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Extending electro-optic bandwidth of native silicon photonic devices well beyond 100 GHz remains a challenge. We demonstrate a scalable C-band silicon photonic platform monolithically integrating ultra-high speed germanium–silicon electro absorption modulators and fin photodiodes. We apply a delta-layer doping process to incorporate Si into Ge. Two electro-absorption modulator variants with lengths of 40 μm and 20 μm exhibit electro optic 3-dB bandwidths of 100 GHz and well beyond 110 GHz, respectively. Co-integrated GeSi-fin photodiodes achieve 3-dB bandwidths of more than 200 GHz. Involving only platform integrated absorption modulators and fin photodiodes, we demonstrate open-eyes in a large signal experiment at 140 Gbit/s. This work establishes an important milestone for next generation optical communication technology.

Integrated electro-optic platforms on silicon (frequently referred to as silicon photonics) constitute an enabling technology for advancing high-performance computing or optical datacenter interconnects due to manufacturing scalability and compatibility with state-of-the-art co-integration or co-packaging with high-speed electronics. The importance of such platforms goes however, well beyond digital optical communications. For example, the anticipated 6 G radio access spectrum¹ now extends up to 300 GHz, for which frequency synthesizers based on integrated photonics are highly relevant due to large frequency scalability, immunity to electro-magnetic interference and low noise. Automotive radar and similar sensing applications considerably benefit in terms of angular resolution^{2,3} from optical signal generation and distribution. Achieving low jitter as a major performance metric in such photonic signal distribution schemes necessitates very large bandwidth platform components such as modulators and photodiodes.⁴ In general, the disruptive potential of optical ultra-broadband signal generation and detection directly leverages on the RF performance of electro-optic platform components.⁵ In this regard, recent years have seen tremendous progress based on bespoke hetero-integration material platforms such as e.g. lithium niobate on silicon⁶ or III-V on silicon.⁷ However, these platforms still pose serious

challenges with respect to manufacturability in general as well as fabrication in a CMOS environment. On the other hand, CMOS compatible studies so far largely concentrated on stand-alone device demonstrations.^{8–13} Full silicon photonic platforms distinctly lack behind in terms of electro-optic RF performance until today. Approaching 100 GBaud signaling has proven very challenging due to the pre-dominant use of free-carrier dispersion in silicon waveguides for optical modulation. The low-efficiency of this effect is a serious bottleneck for further silicon photonic modulator optimization, especially in regard of energy efficiency. With this study, we would like to support a fresh view on CMOS-compatible photonics platforms. The hitherto bandwidth limitations of these platforms may be overcome by chiefly focusing on germanium-silicon technology (GeSi) for electro-optic as well as opto-electric conversion devices. This will limit the platforms to a certain wavelength band (e.g. C-band), but substantial improvements are to be expected in terms of electro-optic bandwidth and energy efficiency. Many applications will actually not require the realization of devices for multiple optical bands. Past demonstrations of GeSi silicon photonics platforms have still been confined to a bandwidth range (<90 GHz) similar to that of carrier dispersion modulators.¹⁴ Here we demonstrate C-band GeSi electro-absorption

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modulators and p-i-n photodiodes fabricated on the same wafer with substantially improved bandwidth reaching up to approximately 200 GHz. The fabrication is done in a 200 mm pilotline with parallel manufacturing of high-speed BiCMOS technologies, i.e. in a CMOS environment with 248 nm deep-UV lithography. The improved performance in comparison to previous attempts on GeSi silicon photonics can be attributed to the use of the germanium-fin concept which permits strong suppression of undesired dopant diffusion in hetero-epitaxial germanium.

Results and discussion

Platform concept and fabrication

To enable a silicon photonic platform for modulation and detection at bandwidths well beyond 100 GHz, we developed a fabrication flow to monolithically integrate GeSi electro-absorption modulators (EAMs) together with GeSi-fin photodiodes (PDs). Efficient optical coupling between standard single-mode fibers and the photonic integrated circuit is achieved through one-dimensional silicon grating couplers, designed to minimize insertion loss and maximize alignment tolerance. For intra-chip optical signal routing, we employ both rib and strip silicon waveguides. Electrical connection is realized with a state-of-the-art back-end-of-line process consisting of five AlCu metal layers.

Both the electro-absorption modulators and the photodiodes were fabricated in a process flow similar to the ultra-fast Ge-fin photodiodes, presented in^{8,15}. The fabrication was entirely conducted on IHP's 200 mm BiCMOS pilot line facilities. In this process, a rather broad germanium region of 400 nm thickness, selectively grown on a silicon waveguide in an SiO₂ window, is subsequently patterned which eventually yields a narrow Ge-fin that is sandwiched between complementary doped Si region. It is worth to note that this approach allows for the simultaneous fabrication of Ge-fins with arbitrary width, which is a difference to vertical devices, where epitaxy-thickness defines the geometry of the intrinsic region. Thus, a broad variety of bandwidth-responsivity combinations can be realized with the exact same process flow, side by side.¹⁶

The application of in-situ doped layers enables very precise control of the desired doping profiles compared to ion implantation. At the same time this approach prevents from unintentional doping of the light-absorbing germanium region which is very beneficial for the high-speed performance of the photodiode, as it efficiently suppresses slow minority carrier diffusion effects. The in-situ doped silicon layers adjacent to the germanium fin also form lateral extensions that enable, after the formation of a nickel-platinum-silicide layer, low-resistance contacts for accessing the photodiode. Thus, the tungsten contacts can be kept at reasonable distance, a prerequisite for maintaining sufficient responsivities. Silicon-nitride spacers prevent from silicidation of the vertical poly-silicon sidewalls in order to avoid extensive optical losses.

While the basic principle of the fabrication approach remains similar to that in⁸, several crucial adjustments were made to enable ultra-fast GeSi electro-absorption modulators with sufficient performance in terms of insertion loss and extinction ratio:

- Adjustment of length and width of the device to improve the extinction ratio and insertion loss
- Local etch of the silicon waveguide to improve coupling into the Ge or GeSi region and enhance mode confinement
- Steeper GeSi sidewalls for low-loss propagation
- Optimization of doping profile in the in-situ doped Si regions to reduce optical losses while maintaining low series resistance
- Replacement of pure Ge by GeSi to shift modulation towards C-band wavelengths

In Fig. 1a, b we show TEM cross-section images of the Ge-fin photodiode from ref. 8 and the GeSi electro-absorption modulator, displaying the adaptations for EAM functionality.

Bulk GeSi electro-absorption modulators operate based on the Franz-Keldysh effect, which causes an increase in a semiconductor's optical absorption coefficient when an electric field is applied. By switching the electric field on or off, the modulator either absorbs the incoming light or allows it to pass through, thereby enabling amplitude modulation. Since the electro-optic bandwidth of Franz-Keldysh effect-based EAMs is not limited by carrier transport times, but by the RC time constant of the device, the width of the device is not a major limiting factor for the modulation speed. For the EAMs, the Ge/GeSi width involves a trade-off between extinction ratio and insertion loss. Modulation efficiency is governed by mode confinement and the electric field strength within the intrinsic zone, whereas insertion loss is primarily due to coupling losses and free-carrier absorption in the doped poly-silicon and silicide regions. We selected a Ge/GeSi width of 400 nm for our EAM to achieve an optimal trade-off between extinction ratio and insertion loss. In the Supplementary Figs. 1, 2 and 3 we provide simulation results confirming that a Ge width of 400 nm yields the most favorable performance.

In order to minimize coupling losses between the silicon waveguide and the Ge/GeSi region and to enhance mode confinement within the EAM, we reduced the thickness of the underlying silicon region from 220 nm to as little as 30 nm via dry etching prior to Ge/GeSi epitaxy. This measure not only decreases the insertion loss and improves the extinction ratio of the EAM but also benefits the concurrently fabricated photodiodes in terms of responsivity¹⁷. In Fig. 1c a TEM image of a longitudinal cut through a GeSi EAM is shown to visualize the butt-coupling approach.

A measure to further prevent from substantial optical losses is the reduction of the doping level in the in-situ doped Si regions that are adjacent to the Ge/GeSi. In contrast to the ultra-fast Ge-fin PDs in ref. 8 we reduced the doping concentration to $1 \times 10^{18} \text{ cm}^{-3}$ in the lower 150 nm of the n- and p-doped poly-silicon layers. In order to ensure a sufficiently low contact resistance a 50 nm silicon layer with a higher doping concentration of $>5 \times 10^{19} \text{ cm}^{-3}$ is added.

One of the most significant modifications is the introduction of GeSi instead of pure Ge. EAMs made from pure Ge show maximum extinction typically at wavelengths beyond 1600 nm, in the L-band wavelength region^{9,18,19}. Although some L-band Ge EAMs demonstrate high speeds of 100 GHz and beyond, their restriction to the L-band significantly limits their applicability across a wider range of applications. By the introduction of around 1% Si into the Ge-region, the bandgap energy at the direct transition in the semiconductor increases enough to shift the operational wavelength region towards the C-band. So far this has been achieved by growing GeSi with a monosilane flow during epitaxy^{20,21}. However, for the incorporation of only about 1% Si in a GeSi layer, the gas flow of silane (SiH₄) or dichlorosilane (SiH₂Cl₂) must remain much lower than the germane (GeH₄) gas flow. Such small levels are particularly difficult to control at chemical vapor deposition (CVD) epitaxy tools which are optimized for the manufacturing of SiGe layers for heterojunction bipolar transistors (HBTs). Further, the conventional approach (a low SiH₄ or SiH₂Cl₂ flow during epitaxy) is prone to loading effects, making it very difficult to control the Si-content, especially for various device sizes and structure densities²².

To circumvent these difficulties, we developed a novel silicon delta doping approach for the epitaxial growth of GeSi layers that fulfill the requirements for C-band EAMs. To the best of our knowledge, such an approach has not been used for GeSi epitaxy so far.

We incorporate 16 silicon delta layers, each comprising two monolayers, introduced at equidistant intervals during the selective epitaxial growth of germanium. Subsequent annealing enables the intermixing of silicon from the delta layers with the surrounding germanium layers. By this, we intend to achieve a nearly homogeneous Si distribution in the Ge body.

This approach decouples the silicon doping process from the germanium growth process, allowing for very precise control of the Si

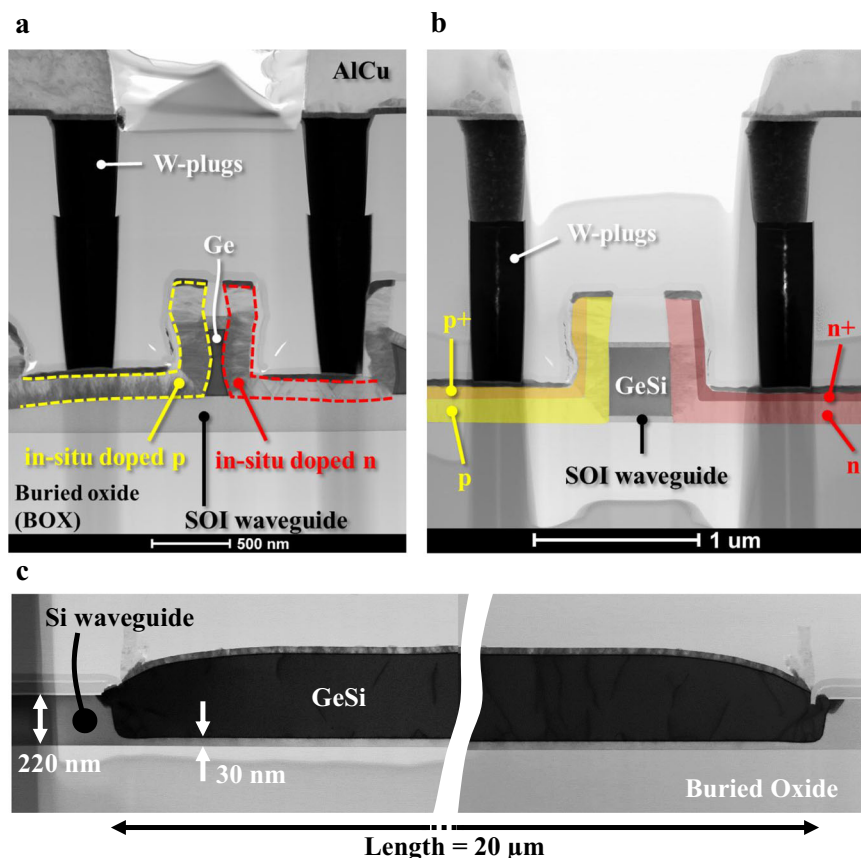


Fig. 1 | TEM images of Ge-fin photodiode and GeSi electro-absorption modulator. **a** Ge-fin PD that features highly in situ-doped contact regions, from ref. 8. **b** GeSi EAM with indicated doping regions with concentrations of $n = p = 1 \times 10^{18} \text{ cm}^{-3}$ and $n^+ = p^+ = 5 \times 10^{19} \text{ cm}^{-3}$. Both cross-sections are cut perpendicular to the direction

of light incidence. **c** TEM longitudinal cut through the germanium-silicon electro-absorption modulator and Si-waveguide to visualize the butt-coupling approach. The in- and outgoing silicon waveguide has a height of 220 nm, while the silicon below the GeSi region is thinned down to 30 nm.

concentration in the germanium. The silicon dose is predominantly governed by the adsorption/desorption equilibrium of silicon on germanium, making this method less sensitive to variations in pattern size and structure density.

All these modifications to the initial Ge-fin photodiode design and fabrication process are intended to enable high-performance GeSi EAMs. However, an additional objective was to simultaneously manufacture ultra-fast GeSi photodiodes.

Since the opto-electrical bandwidth of the photodiode is, amongst other factors, determined by photo-generated carrier transit time, the Ge/GeSi width of the photodiode had to be narrower compared to the EAM. We therefore fabricated PDs with GeSi-fin widths as low as 100 nm to enable ultra-fast detection beyond 200 GHz bandwidth, as well as PDs with widths up to 300 nm, allowing for higher responsivities and bandwidths beyond 110 GHz. In Fig. 2 we provide a schematic process flow that illustrates the fabrication of the GeSi electro-absorption modulators and GeSi photodiodes.

Characterization of the Ge and GeSi electro-absorption modulators

From un-biased and biased transmission measurements we determined insertion losses and static extinction ratios, respectively. Measurements were carried out at wavelengths from 1530 to 1650 nm for both pure Ge as well as GeSi EAMs, each with lengths of 20 μm and 40 μm , as shown in Fig. 3a,b. For the static measurements, the optical power at the fiber tip was set to -3 dBm . We deliberately kept the incident optical power low to minimize self-heating effects on the static extinction ratio. As demonstrated in ref. 23 high optical power in Ge EAMs can cause significant self-heating, resulting in an increased

static extinction ratio that does not translate to a higher dynamic extinction ratio.

The total insertion loss of an electro-absorption modulator is primarily determined by two factors: the coupling losses between the silicon waveguides and the Ge/GeSi active region and the propagation loss within the EAM itself. The propagation loss arises mainly from free-carrier absorption in the doped and silicided regions, scattering losses and bandgap absorption mechanisms. Specifically, these include direct transitions at the Γ -point and indirect transitions at the L-valley of the bandgap diagram. Within the investigated wavelength range, the insertion loss exhibits a notable wavelength dependence, largely due to the varying contribution of direct bandgap absorption at the Γ -point. For optimal operation, the EAM should be used in a wavelength range where the propagation loss is not dominated by direct bandgap absorption under zero bias, while still providing a sufficient extinction ratio. In our GeSi EAM, the low-loss regime beyond 1600 nm could not be characterized due to the bandwidth limitations of the C-band optimized grating coupler. The characterization of the pure Ge EAM is limited by the wavelength limitation of the equipment beyond 1645 nm. Most dynamic measurements were conducted at a wavelength of 1580 nm. At this wavelength, the insertion loss is about 9 dB for the 20 μm long device and approximately 10 dB for the 40 μm long GeSi EAM. We attribute a significant portion of the insertion loss to coupling inefficiencies between the silicon waveguides and the EAM, primarily caused by imperfect mode matching. We anticipate that introducing a taper structure designed to match the effective refractive indices of the silicon waveguide and the EAM can substantially reduce these coupling losses.

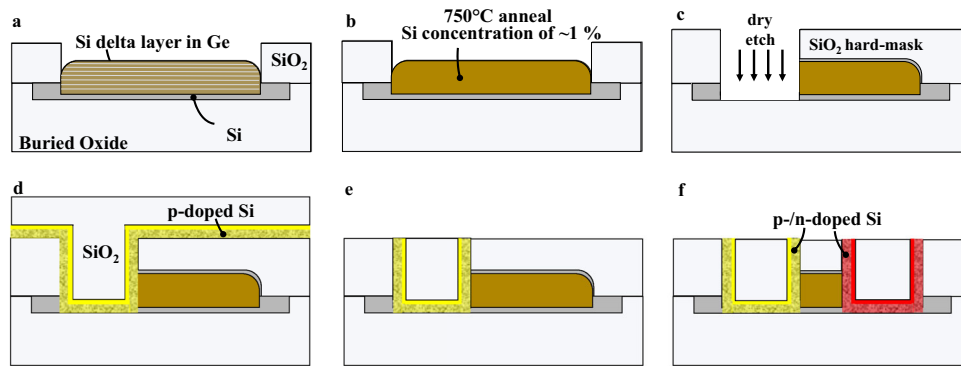


Fig. 2 | Schematic process flow for the GeSi electro-absorption modulator and GeSi photodiode. **a** A broad SOI waveguide region was thinned down to 30 nm thickness. On top of the recessed silicon region a 400 nm thick Ge layer containing 16 silicon delta layers is selectively grown. **b** By applying a 750 °C anneal a nearly homogenous silicon concentration of about 1% is achieved in the Ge body. **c** A silicon-cap is differentially grown and removed on the SiO₂ regions by CMP. Subsequently, a SiO₂ hard-mask is deposited and patterned, followed by a dry-etch of the GeSi and silicon via the hard-mask. **d** An in-situ p-doped silicon is epitaxially grown, followed by a SiO₂ overfill. **e** CMP is performed to restore a planar surface

and to remove the p-doped Si layer in the outer regions. **f** The hard-mask SiO₂ layer is patterned on the opposite side, followed by a GeSi dry etch, in situ n-doped Si epitaxy, plus the planarization steps, as before. The width of the GeSi regions is defined in the layout by the design of the etching mask layers. This enables the simultaneous fabrication of 400 nm wide GeSi EAMs and GeSi-fin photodiodes with active region widths as low as 100 nm. After the formation of the p-i-n structures, the devices are silicided in the contact regions with a nickel-platinum process, followed by the fabrication of the back-end.

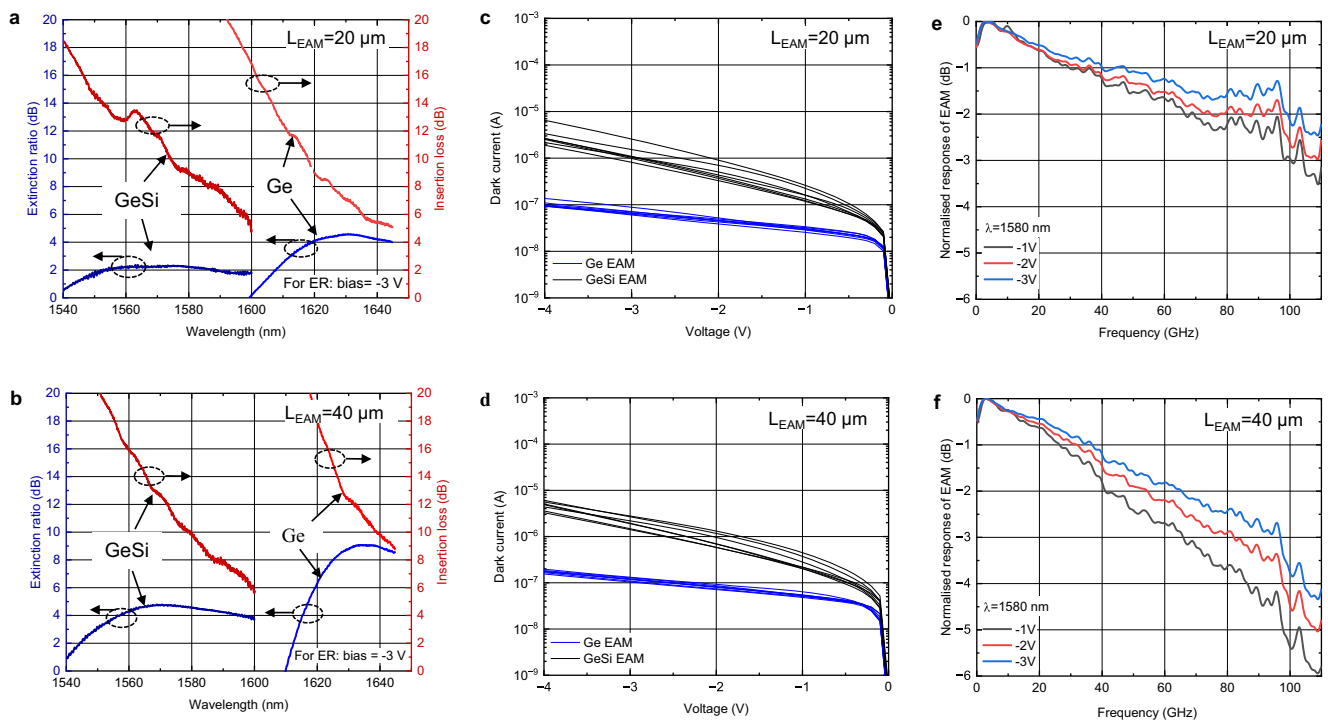


Fig. 3 | Electro-optical and electrical properties of Ge/GeSi EAMs. **a, b** Static extinction ratio (blue curves) at -3 V bias and insertion loss (red curves) of 20 μm long (**a**) and 40 μm long (**b**) Ge/GeSi EAM. The images showcase the shift of the operational wavelengths towards the C-band by the introduction of GeSi over pure Ge. **c, d** Dark currents of 20 μm long (**c**) and 40 μm long (**d**) electro-absorption modulators measured from -4 V to 0 V for Ge and GeSi (7 chips each). **e, f** Normalized frequency response of typical 20 μm long GeSi EAM (**e**) and 40 μm long GeSi EAM (**f**) at λ = 1580 nm for biases from -1 V to -3 V for an optical input power of 2 dBm at fiber tip.

At -3 V bias, the pure Ge EAMs achieve maximum static extinction ratios of around 5 dB at a 20 μm device length and 9 dB at 40 μm length and wavelengths around 1635 nm. These outstanding values were achieved due to the narrow Ge region, enabling high electric field strengths at relatively low applied voltages.

Despite of the same device geometry, the GeSi EAMs yield lower extinction ratios of 2.3 dB at 20 μm and 40 μm at 1575 nm, respectively, which we attribute to an imperfect homogeneity of the silicon within the Ge-region.

It becomes very evident from the measurement results that the GeSi EAMs exhibit a much broader extinction ratio profile across wavelengths compared to the pure Ge devices, which cannot be easily explained under the assumption of a homogeneously distributed silicon concentration in Ge. This is a strong indication of an inhomogeneous silicon distribution: regions with varying silicon content possess different direct bandgap energies. Consequently, while the optical modulation range increases, the maximal extinction ratio decreases. We anticipate that fine-tuning in the GeSi epitaxy process, including

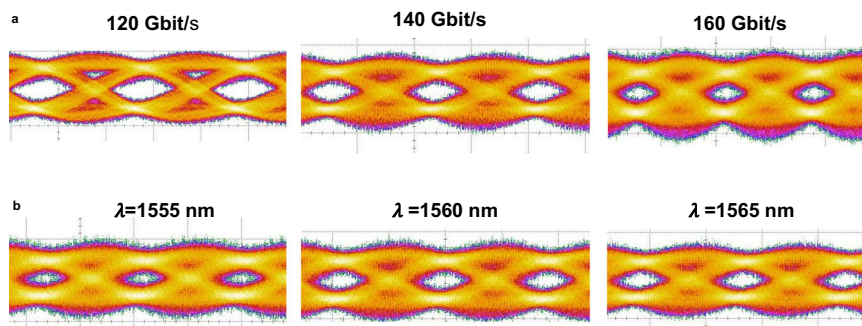


Fig. 4 | Data-transmission experiment at various data rates and wavelengths. **a** Eye diagrams of RF-probed 40 μm long GeSi EAMs at $\lambda=1575$ nm and NRZ-data rates of 120, 140 and 160 Gbit/s. The dynamic extinction ratios are 2.9 dB (120 Gbit/s), 2.25 dB (140 Gbit/s) and 1.6 dB (160 Gbit/s) with 8-tap (120 Gbit/s, 140 Gbit/s) and 16-tap (160 Gbit/s) linear equalization on the receiver side, $V_{\text{bias}} = -1.5$ V and an AWG

output signal of 2.4 V_{pp}. **b** Eye diagrams of RF-probed 40 μm long GeSi EAMs at various C-Band wavelengths and NRZ-data rate of 140 Gbit/s. The dynamic extinction ratios are 1.5 dB (1555 nm), 1.9 dB (1560 nm) and 2.15 dB (1565 nm) with 8-tap linear equalization on the receiver side.

optimized annealing, adjusted layer thicknesses and modified numbers of silicon delta layers, an improved homogeneity and therefore higher extinction ratios that match the extinction ratios of the pure germanium EAMs, can be achieved. Further, an increase of the Si-concentration will shift the operational range even more towards shorter wavelengths, to fully cover the C-band. In ref. 24 it was demonstrated that an optimized, homogenous GeSi alloy enables EAM operation across the entire C-band at low insertion loss, which showcases the suitability of GeSi EAMs for wavelength-division multiplexing applications. Beyond modulation and detection in the C- and L-band a major goal for the future is a silicon photonic platform with bandwidths beyond 100 GHz operating in the O-band, which is widely used for short-reach interconnects within data centers. While we have already demonstrated ultra-fast operation of our Ge-fin photodiodes with 3-dB bandwidths >110 GHz at O-band wavelengths,²⁵ silicon photonic platform compatible O-band modulators have still been lacking behind in bandwidth. Recently, promising results have been demonstrated for quantum-well GeSi electro-absorption modulators operating in the O-band^{26,27}. However, in comparison to bulk GeSi EAMs their fabrication is more complex and pose challenges for seamless integration with ultra-fast photodiodes. Additionally, the electro-optic bandwidths of quantum-well GeSi EAMs have still been confined to bandwidths below 70 GHz. Bulk GeSi EAMs operating in the O-band have not been experimentally demonstrated yet but in ref. 28 it was discussed that O-band modulation is in principle also feasible for Franz-Keldysh effect-based bulk GeSi EAMs through the combination of an increased Si concentration and application of compressive strain onto the GeSi region due to stressor nitrides.

While incorporating Si delta-layers in the Ge body has enabled C-band modulation, this approach impacts the devices' dark current characteristics, as illustrated in Fig. 3c,d. Pure Ge devices exhibit low dark currents of approximately 50 nA at -2 V, whereas GeSi EAMs show an increased dark current of around 800 nA at the same bias. However, we expect that by increasing the number of Si delta layers at a lower dose each combined with optimized annealing conditions, the dark current level is to be further reduced.

Leakage current in EAMs contributes to continuous power consumption, calculated as $P = I_{\text{dark}} \cdot V_{\text{bias}}$. Nevertheless, such dark current levels would contribute to the overall power consumption by only about 1.6 μW , which we consider uncritical.

Dynamic characterization of the GeSi EAMs was performed using a 110 GHz Lightwave Component Analyzer (LCA) from Keysight. Due to the wavelength limitations of the EDFA, which operates only to up approximately 1600 nm we were unable to measure the bandwidth of the pure Ge EAMs at their modulation wavelengths. Consequently, we focused on the dynamic characterization of the GeSi EAMs, that exhibit 3-dB bandwidths of about 100 GHz at -3 V for the 40 μm variant and

even exceed 110 GHz for the 20 μm variant (Fig. 3e,f). We contribute these differences to their series resistances and junction capacitances. From electrical small-signal measurements, we extracted a resistance of 84.8 Ω and a capacitance of 7.88 fF for the 20 μm long device, while for the 40 μm one a resistance of 40.2 Ω and a capacitance of 16.3 fF are obtained. Considering the 50 Ω load of the measurement setup these parameters correspond to theoretical RC-limited 3-dB bandwidths of ~150 GHz and 108 GHz for 20 μm and 40 μm long EAMs, respectively, which are in fairly good agreement with the measured bandwidths. Besides small-signal measurements, we carried out eye-diagram measurements at data rates up to 160 Gbit/s at $\lambda=1575$ nm to demonstrate the large signal behavior of our modulators, provided in Fig. 4a. For the demonstrated transmission experiments we chose the 40 μm long GeSi EAMs due to their higher extinction ratio compared to the 20 μm long devices.

We would like to emphasize here that no pre-equalization was applied on the transmitter side during eye-diagram measurements, but smoothing and minor linear equalization on the oscilloscope (8-tap for 120 Gbit/s and 140 Gbit/s, 16-tap for 160 Gbit/s) at the receiver side. A block diagram of the measurement setup is given in supplementary fig. 4. In the supplementary table 1 we provide SNR and BER values for data rates from 80 Gbit/s to 160 Gbit/s. Across the entire range of investigated data rates, the BER remains below the threshold required for the application of forward error correction (FEC). Furthermore, we obtained 140 Gbit/s eye diagrams at C-Band wavelengths from 1555 nm to 1565 nm, shown in Fig. 4b.

Characterization of GeSi-photodiodes

Our approach is in particular distinguished by the presence of ultra-fast photodiodes being manufactured in conjunction with the EAM. In particular, this enables entire transceiver applications on a single technology platform. Hereby, no separate epitaxy steps for pure Ge and GeSi are to be performed for the photodiode and EAM, respectively. Here we present fin photodiodes with GeSi widths from 100 nm to 300 nm and length of 10 μm , showcasing how the versatile fabrication approach enables a trade-off between RF performance and responsivity.

In Fig. 5a we provide internal responsivity results for various GeSi widths at $\lambda=1550$ nm and $\lambda=1580$ nm at -2 V and -4 V bias. Just like at the electro-absorption modulators the applied voltage leads to an increase in the absorption due to the Franz-Keldysh effect, which can be observed by increased responsivities at these bias conditions. This can also be observed in Fig. 5b, in which the bias dependent internal responsivities are exemplarily shown for the 250 nm wide PD. This device yields a good trade-off between bandwidth and responsivity. Despite the introduction of GeSi the overall internal responsivities are very similar to those from ref. 16 which we attribute to the Franz-

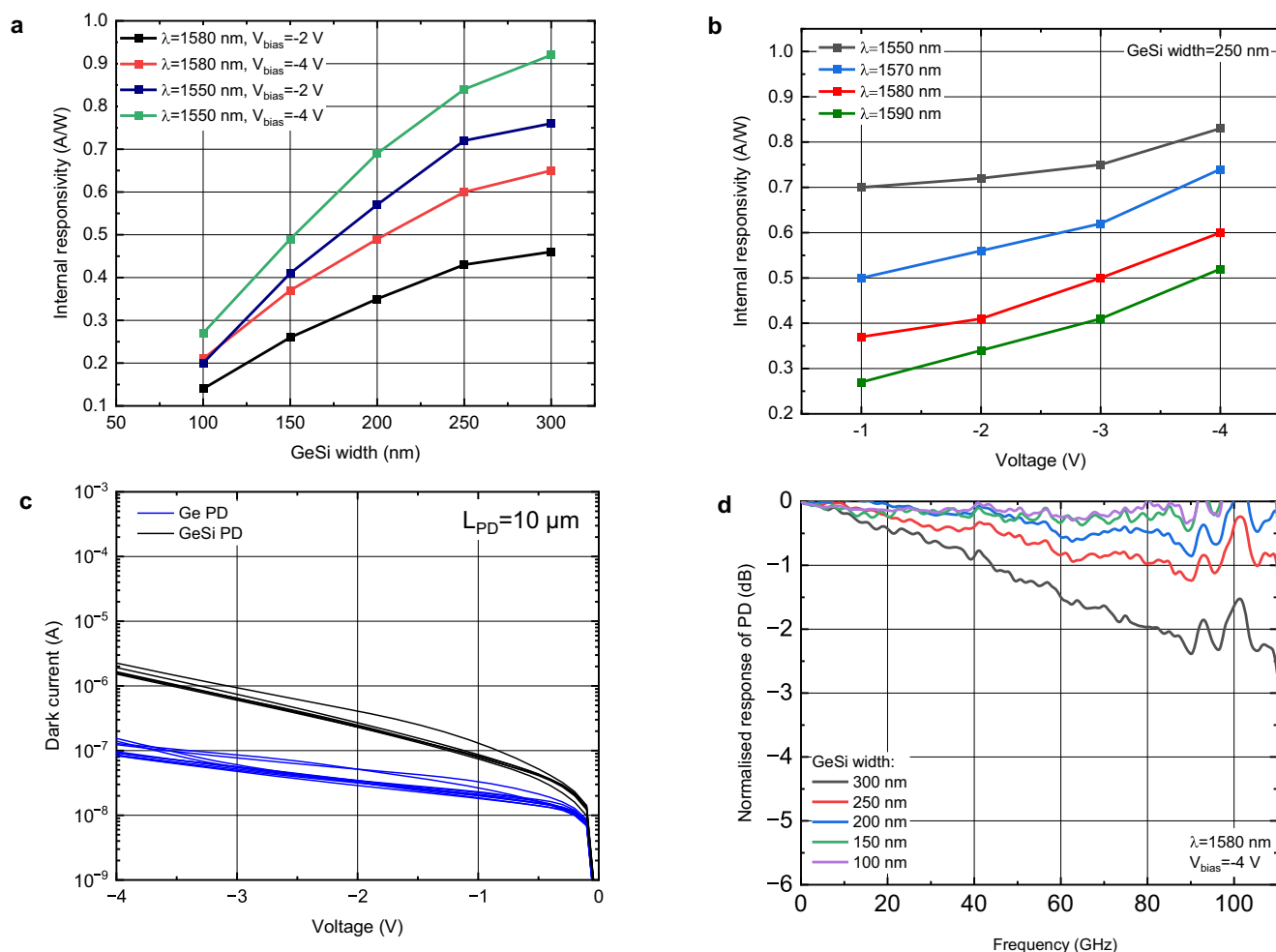


Fig. 5 | Electro-optical and electrical properties of Ge/GeSi PDs. a Internal responsivity vs voltage for PDs with various GeSi widths from 100 nm to 300 nm, measured at $\lambda=1550$ nm and $\lambda=1580$ nm with a bias voltage of -2 V and -4 V. **b** Internal responsivity vs voltage for 250 nm wide GeSi photodiode at wavelengths

of 1550 nm, 1570 nm, 1580 nm and 1590 nm at reverse bias from -1 V to -4 V. **c** Dark currents of 250 nm wide photodiode measured from -4 V to 0 V for Ge and GeSi (7 chips each). **d** Normalized frequency response of PDs with GeSi widths from 100 nm to 300 nm, measured at $\lambda=1580$ nm with -4 V bias.

Keldysh effect, but also the introduction of the local etch of the silicon waveguide prior to the GeSi epitaxy.

Similar to the GeSi EAMs, we notice an increase in the dark current, most probably provoked by the introduction of the Si-delta layers. However, we consider the dark current levels to be still acceptable for state-of-the-art receiver circuits. Due to the shorter length (referred to GeSi EAMs) of $10 \mu\text{m}$ the GeSi PD dark current is around 250 nA at -2 V (Fig. 5c).

Small-signal RF-measurements of the PDs were conducted on a 110 GHz LCA. Due to the setup limitation we were not able to determine the exact 3-dB bandwidth, but for all presented GeSi PD variants the 3-dB bandwidth is well beyond 110 GHz (Fig. 5d). Based on the very flat frequency responses of the narrowest devices of 100 nm and 150 nm width, we expect them to certainly have a 3-dB bandwidth beyond 200 GHz, similar to our Ge-fin photodiodes presented in⁸.

In conclusion, by the introduction of GeSi and all aforementioned adjustments for the sake of the EAM the outstanding PD performance, except for the dark current, remains unaffected.

GeSi-based transceiver

Ultimately, the quality of co-integration of high-speed modulators and detectors in a single platform is characterized by the ability to perform in one system. We therefore compare results from transceiver

experiments from devices realized in Ge or GeSi platforms, by means of EAMs and PDs being actually fabricated in the same process. Very recently, a Ge-based transceiver link has been presented offering Ge EAMs with 67 GHz 3-dB bandwidth and 28 GHz bandwidth Ge PDs, operated at a wavelength of $\lambda=1610$ nm²⁹. The authors show eye diagrams at 32 Gbit/s and 100 Gbit/s transmission without and with digital signal processing (DSP), respectively. In another GeSi-based transceiver experiment, shown in³⁰ a 100 Gbit/s transmission was demonstrated, using a GeSi EAM as modulator as well as photodiode.

In our experiment, performed at $\lambda=1575$ nm, we employ $40 \mu\text{m}$ long GeSi EAMs with a 3-dB bandwidth of 100 GHz and 250 nm wide GeSi-fin PDs with a 3-dB bandwidth well beyond 110 GHz, both fabricated on the same wafer. The applied bias was -1.5 V for the EAM and -4 V for the fin photodiode.

In Fig. 6a,b we provide clearly open eyes at data rates of 120 and 140 Gbit/s, both obtained with only minor linear-equalization on the oscilloscope (8-Tap).

Clearly, this experiment demonstrates significant enhancement of opto-electric device performance of a silicon photonics platform in comparison with the most recent state-of-the-art.

During this experiment, the output signal of the AWG was $2.4 V_{\text{pp}}$. A schematic diagram of the experimental setup is shown in Fig. 6c. A more detailed description of the setup is given in the methods.

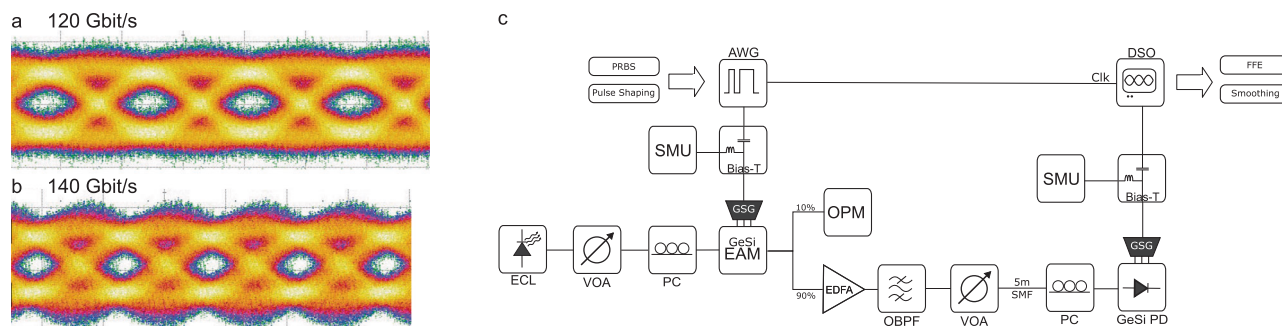


Fig. 6 | Data-transmission experiment GeSi-based transceiver link. a, b Eye diagrams of GeSi-based transceiver link with 100 GHz electro-absorption modulator and >110 GHz fin photodiode, both fabricated on the same wafer. The experiment was conducted at $\lambda = 1575$ nm and NRZ-data rates of 120 Gbit/s and 140

Gbit/s with an output AWG-signal of 2.4 V_{pp}. **c** Block diagram of the experimental setup for high-speed eye diagrams with GeSi EAM on the transmitter side and GeSi fin PD on the receiver side.

Platform benchmark

In order to benchmark our platform against the state-of-the-art in silicon photonics, we compare both modulator and photodiode performance across leading technologies. Specifically, we consider Mach-Zehnder modulators (MZMs), ring modulators and electro-absorption modulators (EAMs), alongside co-integrated germanium (Ge) photodiodes (Table 1).

Best-in-class silicon-photonics platforms to date achieve MZM 3-dB bandwidths of 40–50 GHz, paired with Ge photodiodes reaching up to 89 GHz^{31,32}. More recently, 76 GHz silicon ring modulators have been demonstrated with co-integrated Ge photodiodes operating at 110 GHz, albeit at a modest photocurrent of 200 μ A (dropping to 50 GHz bandwidth at 1 mA)³³. Ge/GeSi EAMs have also shown promise, with estimated 3-dB bandwidths of ~85 GHz at C-band wavelengths. However, their co-integrated photodiodes that are so far limited to 60 GHz have constrained aggregate transceiver performance^{14,34}.

In contrast, our platform achieves an unprecedented combination of ultrafast GeSi EAMs and GeSi photodiodes: the former exhibits a 3-dB bandwidth in excess of 110 GHz, while the latter is estimated to exceed 200 GHz. At C-band wavelengths, both the EAM and the fin-photodiode outperform all previously reported silicon-photonics devices. Moreover, we obtain clearly open eyes at data rates up to 160 Gbit/s, underscoring the exceptional electro-optical performance of our integrated components.

We successfully adapted the Ge-fin fabrication process for the best-in-class photodiodes to enable electro-absorption modulator (EAM) functionality, achieving high-speed modulation with bandwidths exceeding 110 GHz. We apply, for the first time, a delta-layer doping process to incorporate Si into Ge. By introducing GeSi epitaxial growth instead of pure Ge, we extend the application of the EAMs from the L- to C-band wavelengths. Crucially, all modifications in design and fabrication did not compromise the RF performance of the fin-photodiodes, ensuring that their high-speed capabilities remain intact.

This novel process combines by monolithic integration, GeSi EAMs operating at C- and L-band wavelengths with a 3-dB bandwidth of >110 GHz and ultra-fast GeSi-fin photodiodes with a 3-dB bandwidth of >200 GHz, entirely fabricated in IHP's SiGe-BiCMOS pilot line. In terms of bandwidth this outperforms all other silicon photonic devices at C-band wavelengths so far.

In a transmission link experiment integrating a GeSi EAM and photodiode fabricated on the same wafer, we demonstrate high-speed data transmission without the need for extensive digital signal processing.

Our process enables a scalable, energy-efficient silicon photonic platform targeting symbol rates of >200 GBaud. These advances lay the foundation for ultra-high-throughput, low-power optical interconnects essential to the continued scaling of next generation digital

optical communication and ultra-broadband optical signal generation and detection.

Methods

Electro-absorption modulator transmission-measurement

To calculate the extinction ratio and insertion loss of the EAM, the transmission measurements under various bias points were performed. The FormFactor silicon photonics testing platform, based on fully-automated wafer prober CM300xi and automated 6-axis positioning stages (PI), was used. The tunable laser source (Keysight, N777C), optical power meter (Keysight, N7745C) and source/measure unit (SMU) (Keysight, B2912A) were used to perform the measurements. The light from a cleaved SMF (single mode fiber) was coupled in and out via grating couplers using the automated fiber alignment routine. The polarization was manually optimized. The wavelength was swept with a step of 50 pm and transmission spectra were captured. The measurements were repeated for various reverse bias applied to the EAM structure via DC probe. The chuck temperature of the probe station was set to 27 °C.

Photodiode responsivity-measurement

To estimate the photodiode responsivity, the photocurrent was measured at different optical power as function of bias voltage. The FormFactor silicon photonics testing platform, based on fully-automated wafer prober CM300xi and automated 6-axis positioning stages (PI) was used. The tunable laser source (Keysight, N777C) and source/measure unit (SMU) (Keysight, B2912A) were used to perform the measurements. The light from a cleaved SMF was coupled to the grating coupler using the automated fiber alignment routine. The polarization was manually optimized. The excitation wavelength was set to 1550 nm/1570 nm/1580 nm/1590 nm and the power, measured at the fiber tip, was swept from -2.5 dBm to 0 dBm with step 0.5 dB. The characteristics of photocurrent vs. applied voltage in a range from 0.3 V to -4 V with step 0.1 V were obtained. DC probing was performed by standard ground-signal (GS) probe. The quotient of the photocurrent and the optical power (at the fiber tip) at the desire voltage gives the value of external responsivity. Reducing the optical power by subtracting the loss of the grating coupler (measured on 2 mm long waveguide) gives the internal responsivity. The chuck temperature of the probe station was set to 27 °C.

Direct current (D.C.) electrical measurements

The I-V characteristics were measured at wafer level on an automated 200 mm wafer prober (Cascade Microtech) with a temperature-controlled chuck using a DC-analyser (Agilent 4156 C precision semiconductor parameter analyser). An SMU is used to apply the bias (sweep) and ground to the pads of the electro-absorption modulator

Table 1 | Benchmark of state-of-the-art silicon photonic platforms by means of the modulator and photodiode performance

Reference	PETRA ³⁴ 2023	TSMC ³⁵ 2024	AMF ³¹ 2024	imec ¹⁴ 2016	CEA-Lett ³² 2021	IHP (this work)
Modulator type	EAM	Ring-Mod.	MZM	EAM	MZM	EAM
Modulator						
Length or radius(μm)	40	5	-	40	3000	20
Mod. 3-dB BW (GHz)	>67 (= 85)	76	>50	>50 (= 85)	40	>110 (= 140)
Loss (dB)	5	4	n/a	4.5	3	9 (λ = 1580 nm)
Dyn. ER at given data rate (dB) (Mod. Voltage)	1.5 (0.8 Vpp)	n/a	n/a	3 (2 Vpp)	n/a	-
Data rate (NRZ, Gbps)	100	n/a	n/a	50	n/a	-
Photodiode						
PD 3-dB BW (GHz)	60	110	78 (O-band) 89 (C-band)	>50	35	~130 ^b (300 nm width) >200 ^b (100 nm width)
Int. Resp. (A/W) (bias, λ)	0.83 (-4 V, 1550 nm)	1 (O-Band)	1	0.86 ^a (-1 V, 1550 nm)	0.75	0.92 (300 nm width) 0.27 (100 nm width) (-4 V, 1550 nm)

n/a not available in original publication
^a Responsivity is only provided for Ge photodiode, not for GeSi PD, that is co-integrated with GeSi EAM.
^b Estimated value as LCA-measurements are limited to 110 GHz. The flat frequency response of the PD with 100 nm GeSi width is very similar to those of our Ge-photodiodes, yielding a 3-dB bandwidth of 265 GHz (measured with heterodyne measurements)⁸.

or photodiode, respectively. Probing was performed using standard DC probes (tungsten).

S-parameter measurements

The LCA system (Keysight, N5227A) was used to measure the frequency response (S-parameter) of the photodiode (in OE mode) and the EAM modulator (in EO mode). The measurements were performed in a range from 100 MHz to 110 GHz in 50 MHz step. The characteristics were obtained with 8-fold averaging and smoothing applied. S-parameter calibration was done with the impedance standard substrate ISS FormFactor 104-783. The chuck temperature of the probe station (Cascade AP300) was set to 27 °C.

During photodiode characterization an external laser source (Keysight, 81608 A) at 1580 nm was connected to the transmitter of the LCA system. The modulated signal was fed to the photo-diode via manual polarization controller and coupled through grating coupler. The reverse bias voltage was supplied by source/measure unit (Keysight, B2912A) through the PNA's integrated bias-T and 110 GHz RF probe (FormFactor, GSG infinity probe). At the same time the RF response (S₂₁) was measured.

During the EAM measurements, the light from laser source (Keysight, 81608 A) at 1580 nm was coupled to the DUT through the manual polarization controller. The RF input signal as well as DC bias were applied through 110 GHz RF probe (FormFactor, GSG infinity probe). The out-coupled optical signal was amplified by EDFA (FiberLabs AMP-FL8021-CLB) and sent to the LCA-Receiver, where the frequency response was measured.

Eye diagram measurements

Setup with commercial photodiode. The GeSi-EAMs were characterized at wafer level using a Cascade Microtech semiautomated Probe Station AP300, utilizing a 110-GHz RF probe (FormFactor GSG Infinity Probe) and two single fibers for optical coupling. The coupling angle was optimized for 1580 nm. All measurements were carried out at room temperature (with chuck temperature at 27 °C).

Electrical non-return-to-zero (NRZ) signals with a nominal peak-to-peak voltage of 2.4 V and a pseudorandom binary sequence (PRBS) of length 2⁷-1 were generated at various baud rates by a Keysight 8199B arbitrary waveform generator (AWG). A 110-GHz bias tee (SHF BT110R) placed between the AWG and the RF probe provided the reverse bias from a Keysight B2912A source/measure unit (SMU). The optical signal was provided by the external cavity laser source (Keysight 81608 A).

Ten percent of the modulated signal was routed to an optical power meter (Keysight 81634B) to monitor coupling power, while the remaining 90% was amplified by an erbium-doped fiber amplifier (FiberLabs AMP-FL8021-CLB) and subsequently filtered through an optical band-pass filter (EXFO XTA50/W) with a 1-nm bandwidth to eliminate undesired noise.

Then the signal was sent to 100 GHz commercial photodetector (u²t XPDV4121R) connected to a sampling oscilloscope (Keysight DCAX N1046A).

A block diagram of the setup is given in the supplementary file.

We would like to emphasize here that no pre-equalization was applied on the transmitter side during eye-diagram measurements, but smoothing and minor linear equalization on the oscilloscope (8-Tap for 120 G and 140 G, 16-tap for 160 G) at the receiver side.

GeSi-based setup. The setup using the commercial u²t Photodiode was modified. After the optical filter, the signal was sent through 5 m SMF fiber and polarization controller to the GeSi photodiode fabricated at IHP in the same process flow as the GeSi EAM. The probing was done on chip level. The fiber was manually aligned to the grating coupler. The reverse bias voltage was supplied by source/measure unit (Keysight, B2912A) through 110 GHz bias-T (SHF BT110R) and 110 GHz

RF probe (FormFactor, GSG infinity probe). At the same time the eye diagrams were captured on the sampling scope (Keysight, DCAX NI046A). As previously, a linear equalization on the oscilloscope was used.

Data availability

The experimental data and supporting analyses are included in the article and the Supplementary Information. Additional datasets are available from the corresponding authors upon request.

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Author contributions

D.S., S.L. designed and fabricated the electro-absorption modulators and photodiodes and wrote the manuscript with contributions from all authors. Y.Y., W.-C.W. developed the epitaxy processes for the germanium-silicon and in-situ-doped silicon layers. A.P., A.K. performed extensive optoelectrical characterization. J.B. has conducted electro-optical simulations. O.F. has performed ellipsometry and reflectometry measurements during fabrication of the devices. F.B. carried out SIMS-measurements for the development of the GeSi epitaxy process. S.M. developed and conducted the dry etch processes. P.K. optimized and conducted lithography steps. D.W. developed and carried out silicidation processes. D.S., S.L., J.B., L.Z. analysed the results.

Competing interests

The authors declare no competing interests.

Additional information

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